EXPERIMENTS 31-40

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Course code: DSA0503

Subject : Query Processing for Data Science

EXPERIMENT – 31

Aim

To write a Python program to **create a stacked bar plot with error bars**, where the women's bars are stacked on top of the men's bars using the bottom parameter.

Procedure

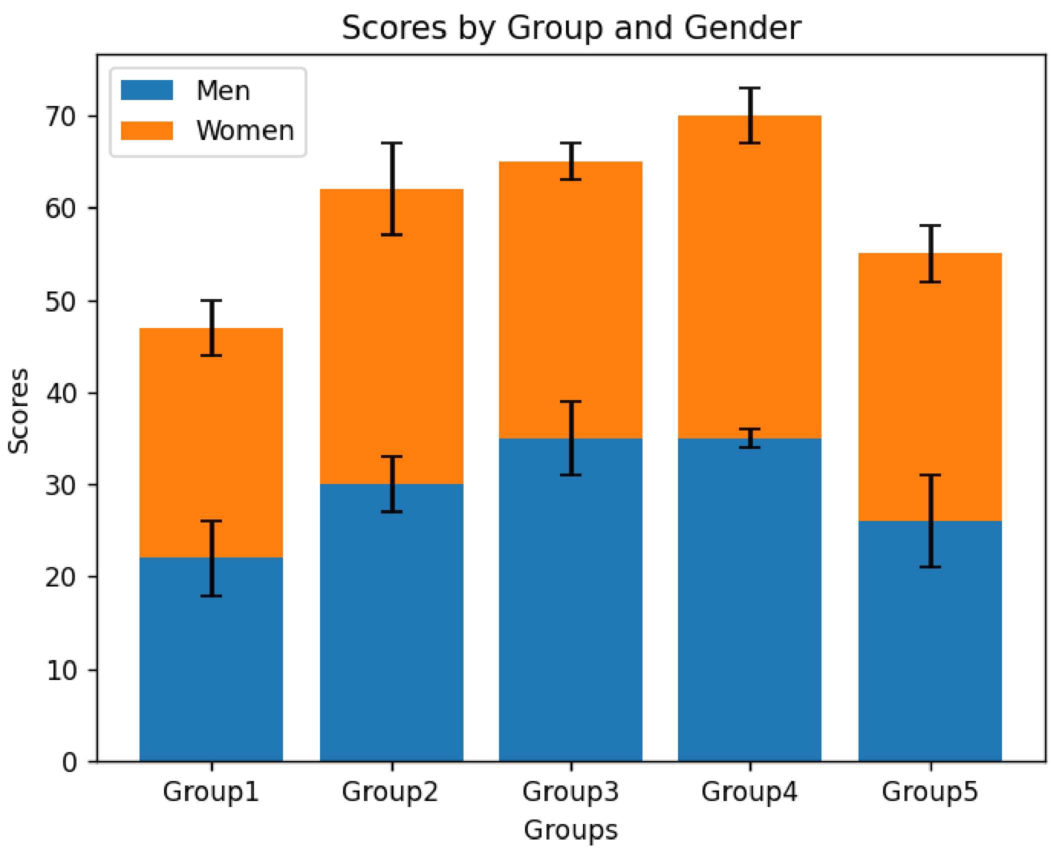
1. Import matplotlib.pyplot.
2. Define the mean scores of men.
3. Define the mean scores of women.
4. Define the standard deviations of men and women.
5. Create the X-axis positions for the five groups.
6. Plot men's scores with error bars.
7. Use bottom=men to stack women's bars above men's bars.
8. Add error bars for women's scores.
9. Add X-axis, Y-axis labels, title and legend.
10. Display the graph.

Code:

```
import numpy as np

import matplotlib.pyplot as plt

groups=['Group1','Group2','Group3','Group4','Group5']

men=np.array([22,30,35,35,26])

women=np.array([25,32,30,35,29])

men_sd=[4,3,4,1,5]

women_sd=[3,5,2,3,3]
```

```
x=np.arange(len(groups))
plt.bar(x,men,yerr=men_sd,capsize=4,label='Men')
plt.bar(x,women,bottom=men,yerr=women_sd,capsize=4,label='Women')
plt.xlabel('Groups')
plt.ylabel('Scores')
plt.title('Scores by Group and Gender')
plt.xticks(x,groups)
plt.legend()
plt.show()
```

Output:



Result

Thus, the Python program to create a stacked bar plot with error bars, with the women's bars stacked on top of the men's bars, was executed successfully.

EXPERIMENT – 32

Aim

To write a Python program to create a horizontal bar chart with differently ordered colors for Language, Science, and Math scores of three students.

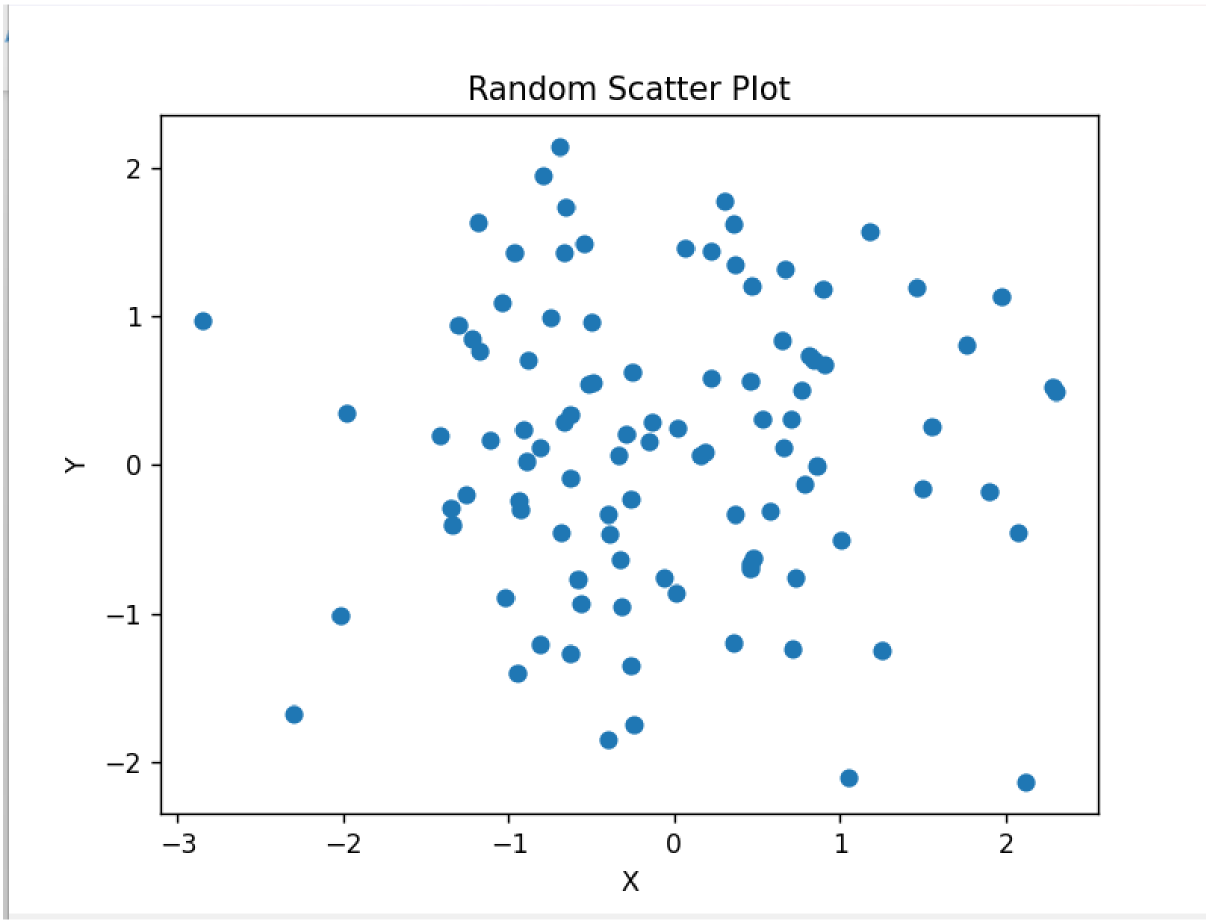
Procedure

1. Import NumPy and Matplotlib.
2. Store the subject order for each student.
3. Store the marks obtained in each subject.
4. Define different colors for the subjects.
5. Arrange the values according to the specified order.
6. Create a horizontal stacked bar chart.
7. Give labels to the students.
8. Add a legend, title and grid.
9. Display the chart.

Code:

```
import numpy as np
import matplotlib.pyplot as plt
x=np.random.randn(100)
y=np.random.randn(100)
plt.scatter(x,y)
plt.xlabel('X')
plt.ylabel('Y')
plt.title('Random Scatter Plot')
plt.show()
```

output:



Result

Thus, the Python program to **create a horizontal bar chart with differently ordered colors** was executed successfully and the required chart was displayed.

EXPERIMENT – 33

Aim

To write a Python program to **generate random X and Y values and draw a scatter plot using empty circles as markers**.

Procedure

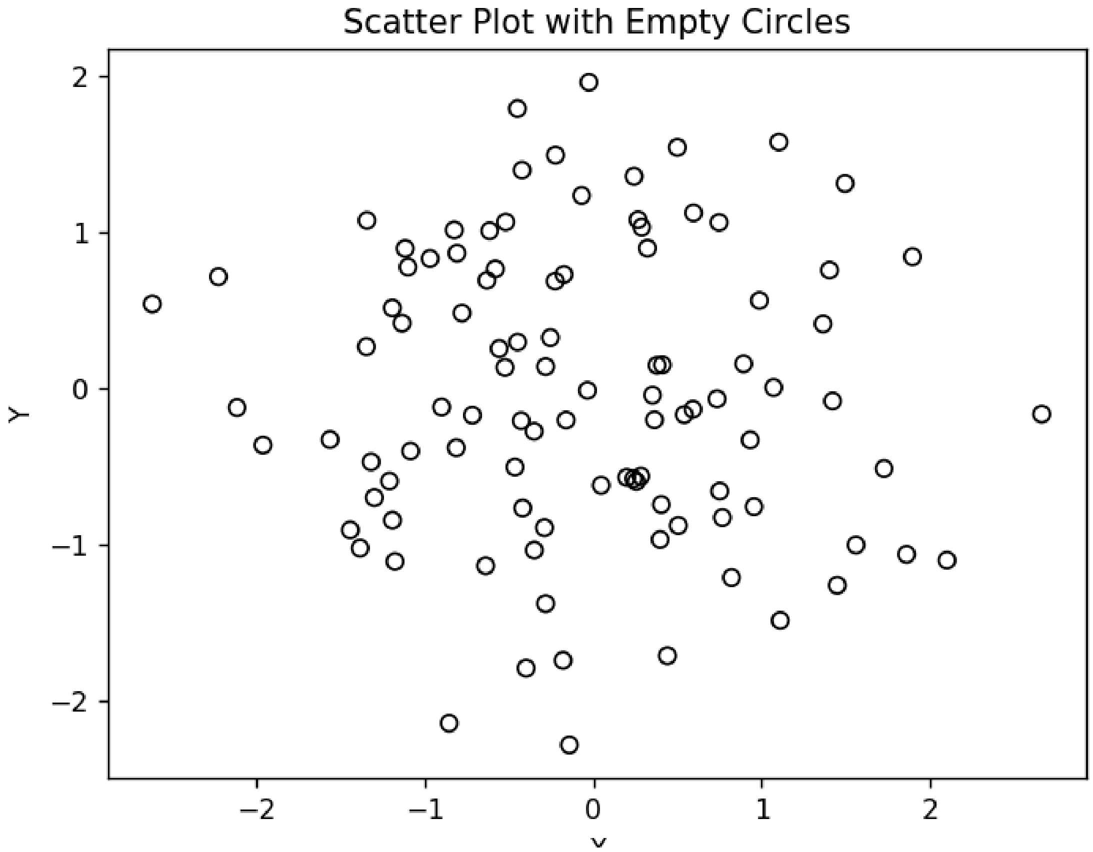
1. Import NumPy and Matplotlib.
2. Generate random values for X.
3. Generate random values for Y.
4. Use plt.scatter() to plot X and Y.
5. Use an empty circle marker.
6. Set the edge color of the circles.
7. Add X-axis and Y-axis labels.

8. Display the scatter plot.

Code:

```
import numpy as np
import matplotlib.pyplot as plt
x=np.random.randn(100)
y=np.random.randn(100)
plt.scatter(x,y,facecolors='none',edgecolors='black')
plt.xlabel('X')
plt.ylabel('Y')
plt.title('Scatter Plot with Empty Circles')
plt.show()
```

output:



Result

Thus, the Python program to draw a scatter plot with empty circles using random distributions in X and Y was executed successfully and the required scatter plot was displayed.

EXPERIMENT – 34

Aim

To write a Python program to draw a scatter plot using randomly generated X and Y values with balls of different sizes.

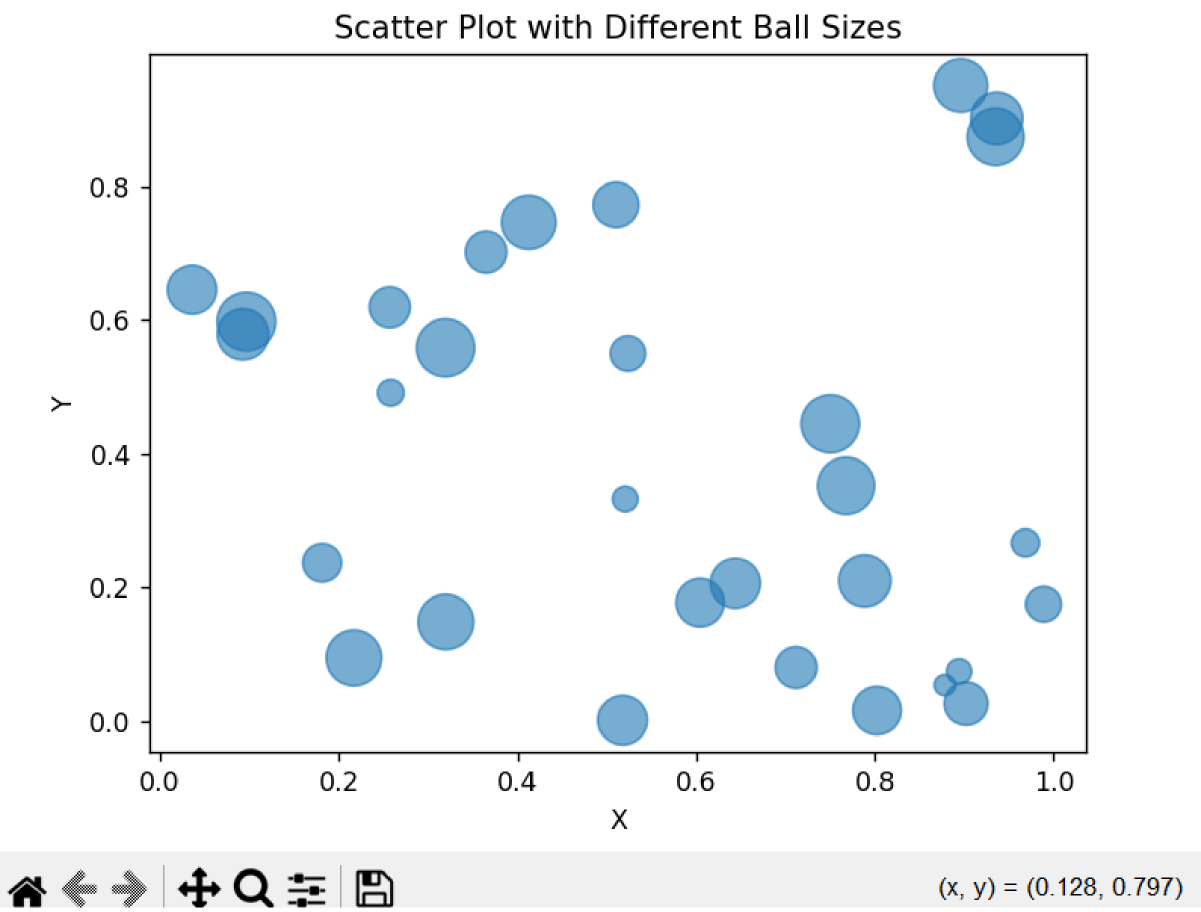
Procedure

1. Import NumPy and Matplotlib.
2. Generate random values for X.
3. Generate random values for Y.
4. Generate random values for the sizes of the balls.
5. Use plt.scatter() to plot the points.
6. Pass the random sizes to the s parameter.
7. Add X-axis and Y-axis labels.
8. Display the scatter plot

Code:

```
import numpy as np

import matplotlib.pyplot as plt

x=np.random.rand(30)

y=np.random.rand(30)

sizes=np.random.randint(50,500,30)

plt.scatter(x,y,s=sizes,alpha=0.6)

plt.xlabel('X')

plt.ylabel('Y')

plt.title('Scatter Plot with Different Ball Sizes')

plt.show()
```

output:



Result

Thus, the Python program to draw a scatter plot using random distributions to generate balls of different sizes was executed successfully and the required scatter plot was displayed.

EXPERIMENT – 35

Aim

To write a Python program to draw a scatter plot comparing the marks obtained in Mathematics and Science by 10 students.

Procedure

1. Import the matplotlib.pyplot library.
2. Store the Mathematics marks of 10 students.
3. Store the Science marks of 10 students.
4. Create a scatter plot using plt.scatter().
5. Label the X-axis as Mathematics Marks.

6. Label the Y-axis as Science Marks.

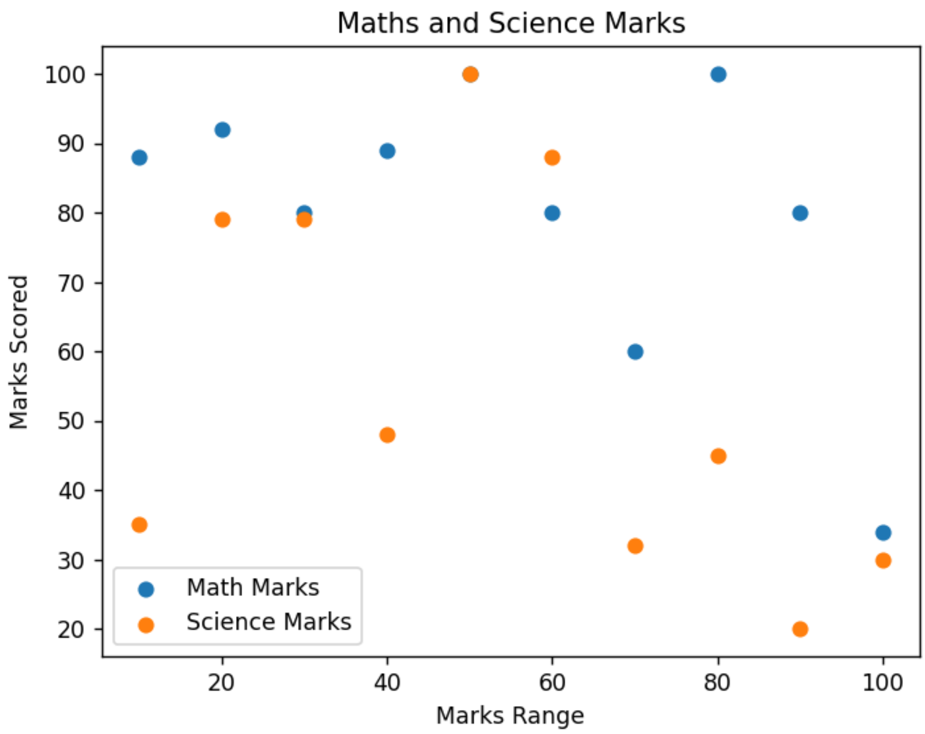
7. Add a suitable title.

8. Display the scatter plot.

Code:

```
import matplotlib.pyplot as plt
math_marks=[88,92,80,89,100,80,60,100,80,34]
science_marks=[35,79,79,48,100,88,32,45,20,30]
marks_range=[10,20,30,40,50,60,70,80,90,100]
plt.scatter(marks_range,math_marks,label='Math Marks')
plt.scatter(marks_range,science_marks,label='Science Marks')
plt.xlabel('Marks Range')
plt.ylabel('Marks Scored')
plt.title('Maths and Science Marks')
plt.legend()
plt.show()
```

output:



Result

Thus, the Python program to draw a scatter plot comparing Mathematics and Science marks of 10 students was executed successfully and the required scatter plot was displayed.

EXPERIMENT – 36

Aim

To write a Python program to draw a scatter plot for three different groups comparing their weights and heights.

Procedure

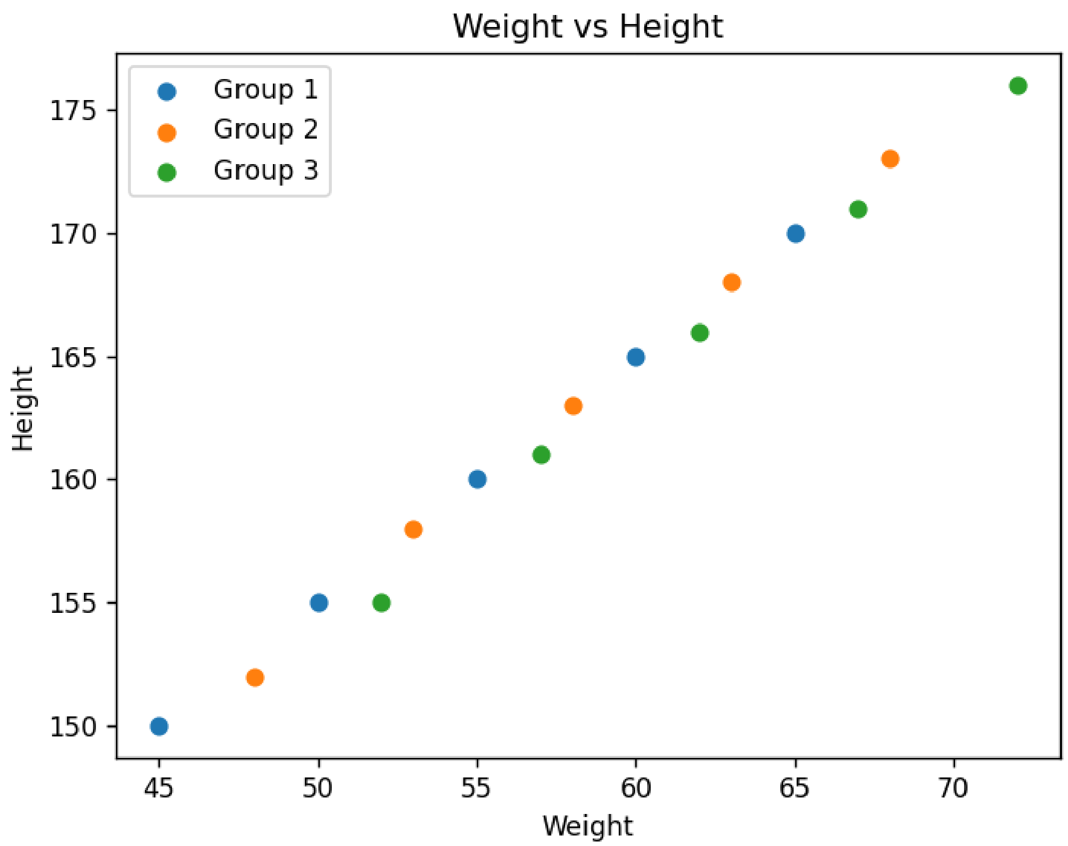
1. Import the matplotlib.pyplot library.
2. Define the weight and height values for three groups.
3. Create a scatter plot for each group.
4. Use different markers or colors to distinguish the groups.
5. Add X-axis and Y-axis labels.
6. Add a suitable title.
7. Add a legend.
8. Display the scatter plot.

Code:

```
import matplotlib.pyplot as plt
weight1=[45,50,55,60,65]
height1=[150,155,160,165,170]
weight2=[48,53,58,63,68]
height2=[152,158,163,168,173]
weight3=[52,57,62,67,72]
height3=[155,161,166,171,176]
plt.scatter(weight1,height1,label='Group 1')
plt.scatter(weight2,height2,label='Group 2')
plt.scatter(weight3,height3,label='Group 3')
plt.xlabel('Weight')
plt.ylabel('Height')
```

```
plt.title('Weight vs Height')
plt.legend()
plt.show()
```

Output:



Result

Thus, the Python program to draw a scatter plot for three different groups comparing weights and heights was executed successfully and the required scatter plot was displayed.

EXPERIMENT – 37

Aim

To write a Python program using Pandas to create a DataFrame from a dictionary and display the DataFrame.

Procedure

1. Import the Pandas library.
2. Create a dictionary containing columns X, Y, and Z.

3. Store five values in each column.

4. Convert the dictionary into a Pandas DataFrame.

5. Display the DataFrame using print().

Code:

```
import pandas as pd
data={'X':[78,85,96,80,86],'Y':[84,94,89,83,86],'Z':[86,97,96,72,83]}
df=pd.DataFrame(data)
print(df)
```

OUTPUT:

```
py
    X   Y   Z
0  78  84  86
1  85  94  97
2  96  89  96
3  80  83  72
4  86  86  83
```

Result

Thus, the Pandas program to create a DataFrame from a dictionary and display it was executed successfully.

EXPERIMENT – 38

Aim

To write a Pandas program to create and display a DataFrame from a specified dictionary with index labels.

Procedure

1. Import the Pandas library.

2. Import NumPy for representing missing values.

3. Create the given dictionary containing name, score, attempts, and qualify.

4. Create the given index labels.

5. Create a DataFrame using pd.DataFrame().

6. Assign the index labels to the DataFrame.

7. Display the DataFrame.

Code:

import pandas as pd

import numpy as np

exam_data={'name':['Anastasia','Dima','Katherine','James','Emily','Michael','Matthew','Laura', 'Kevin','Jonas'],'score':[12.5,9,16.5,np.nan,9,20,14.5,np.nan,8,19],'attempts':[1,3,2,3,2,3,1,1,2, 1],'qualify':['yes','no','yes','no','no','yes','yes','no','no','yes']}

labels=['a','b','c','d','e','f','g','h','i','j']

df=pd.DataFrame(exam_data,index=labels)

print(df)

output:

```
py
        name  score  attempts qualify
a  Anastasia   12.5         1     yes
b       Dima    9.0         3      no
c  Katherine   16.5         2     yes
d      James    NaN         3      no
e      Emily    9.0         2      no
f    Michael   20.0         3     yes
g    Matthew   14.5         1     yes
h      Laura    NaN         1      no
i      Kevin    8.0         2      no
j      Jonas   19.0         1     yes
>>>
```

Result

Thus, the Pandas program to create and display a DataFrame from the specified dictionary with index labels was executed successfully.

EXPERIMENT – 39

Aim

To write a Python program using Pandas to create a DataFrame and display its first three rows.

Procedure

1. Import the Pandas library.
2. Import NumPy to represent missing values.
3. Create the given dictionary data.
4. Create the specified index labels.
5. Create a DataFrame using pd.DataFrame().

6. Use the head(3) function to select the first three rows.
7. Display the result.

Code:

import pandas as pd

import numpy as np

exam_data={'name':['Anastasia','Dima','Katherine','James','Emily','Michael','Matthew','Laura', 'Kevin','Jonas'],'score':[12.5,9,16.5,np.nan,9,20,14.5,np.nan,8,19],'attempts':[1,3,2,3,2,3,1,1,2, 1],'qualify':['yes','no','yes','no','no','yes','yes','no','no','yes']}

labels=['a','b','c','d','e','f','g','h','i','j']

df=pd.DataFrame(exam_data,index=labels)

print(df.head(3))

output:

```
py
        name  score  attempts qualify
a  Anastasia   12.5         1     yes
b       Dima    9.0         3      no
c  Katherine   16.5         2     yes
>>
```

Result

Thus, the Pandas program to get and display the first three rows of the given DataFrame was executed successfully.

EXPERIMENT – 40

Aim

To write a Python program using Pandas to select and display the name and score columns from a given DataFrame.

Procedure

1. Import the Pandas library.
2. Import NumPy for missing values.
3. Create the given dictionary data.
4. Create the index labels.
5. Create a DataFrame using pd.DataFrame().

6. Select the name and score columns.
7. Display the selected columns.

Code:

```
import pandas as pd

import numpy as np

exam_data={'name':['Anastasia','Dima','Katherine','James','Emily','Michael','Matthew','Laura',
'Kevin','Jonas'],'score':[12.5,9,16.5,np.nan,9,20,14.5,np.nan,8,19],'attempts':[1,3,2,3,2,3,1,1,2,
1],'qualify':['yes','no','yes','no','no','yes','yes','no','no','yes']}

labels=['a','b','c','d','e','f','g','h','i','j']

df=pd.DataFrame(exam_data,index=labels)

print(df[['name','score']])
```

output:

```
        name  score
a  Anastasia   12.5
b       Dima    9.0
c  Katherine   16.5
d      James    NaN
e      Emily    9.0
f    Michael   20.0
g    Matthew   14.5
h      Laura    NaN
i      Kevin    8.0
j      Jonas   19.0
>
```

Result

Thus, the Pandas program to select and display the name and score columns from the given DataFrame was executed successfully.